**(14 marks) Choose the correct answer

Q1) One of the following systems is linear

a) $\begin{aligned} x_1 - 2x_2 &= 3 \\ 2x_1 + (x_2)^2 &= 4 \end{aligned}$ b) $\begin{aligned} x_1 - 2x_2 &= \sqrt{2} \\ 2x_1 + (\ln 7)x_2 &= \frac{3}{5} \end{aligned}$ c) $\begin{aligned} x_1 &= 3x_2 - 7 \\ 2x_1 - \frac{1}{x_2} &= \frac{1}{3} \end{aligned}$ d) $\begin{aligned} x_1 - 2x_2 &= 4 \\ 3x_1 + 5x_1x_2 &= 0 \end{aligned}$

Q2) *If A and B are* 3×3 matrices with $\det A = -3$ and $\det B = 8$, then $\det(3B)^t(2A)^{-1} =$

a) $\frac{-64}{81}$ b) -9 c) $\frac{-16}{9}$ d) $\frac{-1}{9}$

Q3) If A and B are 2x2 matrices with trA=5, trB=-3 and trAB=2, then $tr(3BA)^t =$

a) 54 b) -45 c) -135 d) 6

Q4) *If* A is 5×3 matrix and AX=YA=A, then

a) $X = I_3, Y = I_3$ b) $X = I_3, Y = I_5$ c) $X = I_5, Y = I_3$ d) $X = I_5, Y = I_5$

Q5) One of the following expressions makes sense for any vectors u,v, and w in $\Re^n$

a) $\|u.v\|$ b) $\|u\|.\|v\|$ c) $u.(v.w)$ d) $u.v - 3\|u\|$

Q6) If $\det\begin{bmatrix} a & b \\ c & d \end{bmatrix} = 2$, then $\det\begin{bmatrix} 2a & c \\ -6b & -3d \end{bmatrix} =$

a) -72 b) 12 c) -12 d) 72

Q7) *If* A is n×n matrix with $A^2 = I - 4A$, then $A^{-1} =$

a) $A - 4I$ b) $A + 4I$ c) $4A$ d) $-4A$

**(8 marks) Fill in the blanks

Q9) If $A=\begin{bmatrix}3 & 5\\ 3 & d\end{bmatrix}$, $C(A)=\begin{bmatrix}1 & -2\\ C_{12} & C_{22}\end{bmatrix}$, then detA= ....-7.... 4

(Where C(A) is the matrix of cofactors)

Q10) If the linear system $\begin{matrix}x+2y=1\\ 2x+(a^2-5)y=a+5\end{matrix}$ has No solution, then a=....3

Q11) The solution of the augmented matrix $\begin{bmatrix}1 & 2 & 3 & 3 & 3\\ 0 & 0 & 1 & -1 & 1\end{bmatrix}$ is

$(x_1,x_2,x_3,x_4)=$ .....

Q12) Given the linear system $\begin{matrix}ax+by=1\\ cx+dy=2\end{matrix}$, with $\det\begin{bmatrix}a & b\\ c & d\end{bmatrix}=2$, $\det\begin{bmatrix}2 & b\\ 4 & d\end{bmatrix}=8$, $\det\begin{bmatrix}a & -1\\ c & -2\end{bmatrix}=6$,

then x=...1.., y=...1....

** (4 marks)Answer the following questions as True or False

Q13) For any nxn matrix A we have A^tA is always a symmetric matrix. True

Q14) $\det(A-B)=\det A-\det B$ false

Q15) If A is nxn matrix with detA=2 ,then the solution of the homogeneous system $AX=0$ is Only the trivial solution. false

Q16) The sum of two invertible matrices is invertible. True

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Q17) (4 marks) Solve the following linear system in detail using Gauss Jordan elimination

$$x_1 + x_2 + 2x_3 = 8$$
$$x_1 + 2x_2 - 3x_3 = -1$$
$$x_1 + 2x_2 + x_3 = 7$$

Unique Solution

$$\begin{bmatrix} 1 & 1 & 2 & 8 \\ 1 & 2 & -3 & -1 \\ 1 & 2 & 1 & 7 \end{bmatrix}$$

$$\begin{bmatrix} 1$$

